

Re-analysis Summary Report for Peer-evaluation

Paper ID: **Hendricks_QuartJournEco_2018_wNKW**

Re-analyst's ID: **543x6**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:

... immigrants who have never been to high school gain ... more on migration to the United States than immigrants with a college degree (p. 682.)

The corresponding article

Here you can find the original data and article: https://osf.io/ra8qk/?view_only=a863e0709fca4f02807c10b11bb69b05

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

I create variables for schooling and dummies to control for Countries. I create dependent variable: difference between US adjusted wage and home adjusted wage. I regress schooling on wage gains, including dummies to control for the effect of Country. H_0: education level affects negatively wage gains at migration. Results: the wage gain at migration decreases as the number of years of schooling of the migrants increase.

The analyst's conclusion: People who did not attend school (or went to school for less than 9 years in total) gain more at migration in term of wage increase. This is confirmed by the fact that the only significant variables are "no school" and "some high school". In particular, we can also note that the magnitude and the significance of the coefficients increase as the years of school decrease. To deal with Country fixed effects, we remove Peru as it is an outlier (it represents just a few dirty observations) and we create dummy variable for the other Countries. Since the Countries have all the same weights, Mexico is the only significant control variable as its sample size is much greater than that of the other Countries.

The analyst's categorization of the result:

The results show evidence for the null-hypothesis

In Task 2, the analysts received the following instruction to conduct the analysis:

You should use the pooled sample of immigrants from poor countries in the NIS (New Immigrant Survey) and the Migration Projects, including Mexico. You should not control for sector: agriculture/nonagriculture, and region i.e. whether an immigrant grew up in a rural or an urban area.

Here, the analyst reported the following steps and results of the analysis:

I estimate how education level affects wage gains at migration, controlling for Country fixed effects. The dependent variable is the wage gains at migration. First of all, I create variable for schooling and dummies to control for Countries. Then, I create the dependent variable as difference between US adjusted wage and home adjusted wage. Finally, I regress schooling on wage gains, including dummies to control for the effect of Country. My aim is to test whether the wage gain at migration decreases as the number of years of schooling of the migrants increase. The analysis shows that people who did not attend school (or went to school for less than 9 years in total) gain more at migration in terms of wage increase. This is confirmed by the fact that the only significant variables are “no school” and “some high school”. In particular, we can also note that the magnitude and the significance of the coefficients increase as the years of school decrease. To deal with Country fixed effects, we remove Peru as it is an outlier (it represents just a few dirty observations) and we create dummy variable for the other Countries. Since the Countries have all the same weights, Mexico is the only significant control variable as its sample size is much greater than that of the other Countries. I perform a joint Hypothesis testing, using the F-Statistic, to test whether the coefficients related to some high school but no degree (vSHS), high school degree (vHSD) and some college (vSC) are zero. The value of the F-statistic for the joint hypothesis test is approximately equal to 0.4029. Its p-value is approximately equal to $0.759 > 0.05$. This means that we cannot reject the null hypothesis that the coefficients vSHS, vHSD and vSC are zero. Therefore, the final model is the following (controlling for Countries).

Analysis language/software/code (optional to check):

Task 1: R

Task 2: R

Analysis codes: <https://osf.io/fst84/>